THE UK UNIVERSITY INTEGRATION BEE 2024/2025

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ROUND THREE DICHOTOMY SOLUTIONS

Sponsored by


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1 Set 2/Segment 1

1. $\displaystyle\int_0^2 \sqrt{4-x^2}\,\mathrm{d}x$ and $\displaystyle\int_0^{\sqrt{2}} \sqrt{4-x^2}\,\mathrm{d}x$

   Solution. The integrand is an upper half-circle with center the origin. In the first, the integration domain makes it the area of a quarter circle of radius 2, so it's just π.

   More generally, the area is the sum of a sector and a triangle, more precisely

   $$\begin{aligned}\int_0^a \sqrt{4-x^2}\,\mathrm{d}x &= \frac{\sin^{-1}\left(\frac{a}{2}\right)}{2\pi}\cdot 4\pi + \frac{1}{2}\cdot a\cdot\sqrt{4-a^2}\\ &= 2\sin^{-1}\left(\frac{a}{2}\right) + \frac{a\sqrt{4-a^2}}{2}.\end{aligned}$$

   It follows for the second integral that

   $$\int_0^{\sqrt{2}} \sqrt{4-x^2}\,\mathrm{d}x = 2\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) + \frac{\sqrt{2}\sqrt{4-2}}{2} = \frac{\pi}{2}+1.$$

   :)

2. $\displaystyle\int \frac{\tan(2x)}{\tan(x)}\,\mathrm{d}x$ and $\displaystyle\int \frac{\tan(x)}{\tan(2x)}\,\mathrm{d}x$

   Solution. From $\tan(2x) = \dfrac{2\tan(x)}{1-\tan^2(x)}$ the integrals are

   $$\int \frac{2}{1-\tan^2(x)}\,\mathrm{d}x \qquad \text{and} \qquad \int \frac{1-\tan^2(x)}{2}\,\mathrm{d}x.$$

   The second integral is much easier. Using $\tan^2(x) = \sec^2(x) - 1$, we have

   $$\int \frac{1-\tan^2(x)}{2}\,\mathrm{d}x = \int \frac{2-\sec^2(x)}{2}\,\mathrm{d}x = x - \frac{\tan(x)}{2} + C.$$

   For the first integral, let $u = \tan(x)$, then

   $$\begin{aligned}\int \frac{2}{1-\tan^2(x)}\,\mathrm{d}x &= \int \frac{2}{(1-u^2)(1+u^2)}\,\mathrm{d}u\\ &= \int \frac{1}{1-u^2} + \frac{1}{1+u^2}\,\mathrm{d}u\\ &= \int \frac{1}{2(1-u)} + \frac{1}{2(1+u)} + \frac{1}{1+u^2}\,\mathrm{d}u\\ &= \frac{1}{2}\left(\ln|1+u| - \ln|1-u|\right) + \tan^{-1}(u) + C\\ &= \frac{1}{2}\left(\ln|1+\tan(x)| - \ln|1-\tan(x)|\right) + x + C.\end{aligned}$$

   :)

3. $\displaystyle\int_0^\infty \frac{1}{1+e^{\sqrt{x}}}\,\mathrm{d}x$ and $\displaystyle\int_0^\infty \frac{1}{1+(\sqrt{x})^e}\,\mathrm{d}x$

Solution. Both integrals use the same substitution $u = \sqrt{x}$. In the first integral we get

$$\begin{aligned}\int_0^\infty \frac{1}{1+e^{\sqrt{x}}}\,\mathrm{d}x &= \int_0^\infty \frac{2u}{1+e^u}\,\mathrm{d}u \\ &= \int_0^\infty \frac{2ue^{-u}}{1+e^{-u}}\,\mathrm{d}u \\ &= 2\sum_{n=0}^\infty \int_0^\infty ue^{-u}(-e^{-u})^n\,\mathrm{d}u \\ &= 2\sum_{n=0}^\infty (-1)^n \int_0^\infty ue^{-(n+1)u}\,\mathrm{d}u \\ &= 2\sum_{n=0}^\infty \frac{(-1)^n}{(n+1)^2} \\ &= \frac{\pi^2}{6}.\end{aligned}$$

In the second integral we get

$$\int_0^\infty \frac{1}{1+(\sqrt{x})^e}\,\mathrm{d}x = \int_0^\infty \frac{2u}{1+u^e}\,\mathrm{d}u.$$

Let $u = y^{\frac{1}{e}}$, then we get

$$\begin{aligned}\int_0^\infty \frac{2y^{\frac{1}{e}}}{1+y}\cdot\frac{1}{e}y^{\frac{1-e}{e}}\,\mathrm{d}y &= \frac{2}{e}\int_0^\infty \frac{y^{\frac{2}{e}-1}}{1+y}\,\mathrm{d}y \\ &= \frac{2}{e}\,\mathrm{B}\left(\frac{2}{e}, 1-\frac{2}{e}\right) \\ &= \frac{2}{e}\Gamma\left(\frac{2}{e}\right)\Gamma\left(1-\frac{2}{e}\right) \\ &= \frac{2\pi}{e}\csc\left(\frac{2\pi}{e}\right).\end{aligned}$$

:)

4. $\displaystyle\int_0^\infty \sqrt{x^2\sqrt{2^{-x}\sqrt{x^4\sqrt{4^{-x}\sqrt{x^8\sqrt{8^{-x\sqrt{\cdots}}}}}}}}\,\mathrm{d}x$ and $\displaystyle\int_0^{\frac{1}{2}} \frac{x}{\sqrt{x+\frac{x}{\sqrt{x+\ldots}}}} + \sqrt{x^3+x^2\sqrt{x^3+x^2\sqrt{x^3+\ldots}}}\,\mathrm{d}x$

Solution. First integral: Without considering convergence issues (formally, we should be doing a limit, left to the reader), we have

$$\begin{aligned}\sqrt{x^2\sqrt{2^{-x}\sqrt{x^4\sqrt{4^{-x}\sqrt{x^8\sqrt{8^{-x\sqrt{\cdots}}}}}}}} &= x^{\frac{2}{2}}\cdot 2^{-\frac{x}{4}}\cdot x^{\frac{4}{8}}\cdot 2^{-\frac{2x}{16}}\cdot x^{\frac{8}{32}}\cdot 2^{-\frac{3x}{64}}\cdot\ldots \\ &= \prod_{n=1}^\infty x^{\frac{2^n}{2^{2n-1}}}\prod_{m=1}^\infty 2^{-\frac{m}{2^{2m}}x} \\ &= x^{\sum_{n=1}^\infty \frac{1}{2^{n-1}}}2^{-x\sum_{m=1}^\infty \frac{m}{4^m}} \\ &= x^2 2^{-\frac{4}{9}x},\end{aligned}$$

where the last step comes from a simple series evaluation of $\sum mx^m$ in the second term. Then we get

$$\begin{aligned}\int_0^\infty x^2 2^{-\frac{4}{9}x}\,\mathrm{d}x &= \text{IBP twice}\\ &= \frac{729}{32(\ln(2))^3}.\end{aligned}$$

Another way to see this is that it's just a gamma function, particularly $\Gamma(3) = 2$

$$\int_0^\infty x^2 2^{-\frac{4}{9}x}\,\mathrm{d}x = \frac{9^3\Gamma(3)}{4^3(\ln(2))^3} = \frac{729}{32(\ln(2))^3}$$

Second integral: For the first function in the integrand, we have

$$f(x) = \frac{x}{\sqrt{x+f(x)}} \implies f(x)^3 + f(x)^2 x - x^2 = 0$$

and for the second function,

$$g(x) = \sqrt{x^3 + x^2 g(x)} \implies x^3 + x^2 g(x) - g(x)^2 = 0.$$

But this implies $f = g^{-1}$. Since $g(0) = 0$ and $g\left(\frac{1}{2}\right) = \frac{1}{2}$, it follows the integral evaluates to $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. :)

5. $\displaystyle\int_{e^2}^{e^e} \frac{\ln\left(\frac{x^{(\ln(\ln(x)))^2}}{\ln(x)}\right)}{(\ln(\ln(x)))^3 \ln(x)}\,\mathrm{d}x$ and $\displaystyle\int_0^1 \ln\left(x^{\ln(1-\ln(x))+1-\ln(x)}\right)\,\mathrm{d}x$

Solution. First integral is secretly a reverse quotient rule:

$$\begin{aligned}\int_{e^2}^{e^e} \frac{\ln\left(\frac{x^{(\ln(\ln(x)))^2}}{\ln(x)}\right)}{(\ln(\ln(x)))^3 \ln(x)}\,\mathrm{d}x &= \int_{e^2}^{e^e} \frac{(\ln(\ln(x)))^2 \ln(x) - \ln(\ln(x))}{(\ln(\ln(x)))^3 \ln(x)}\,\mathrm{d}x\\ &= \int_{e^2}^{e^e} \frac{\ln(\ln(x)) - \frac{1}{\ln(x)}}{(\ln(\ln(x)))^2}\,\mathrm{d}x\\ &= \left[\frac{x}{\ln(\ln(x))}\right]_{e^2}^{e^e}\\ &= e^e - \frac{e^2}{\ln(2)}.\end{aligned}$$

The second integral is more bashy.

$$\begin{aligned}&\int_0^1 \ln\left(x^{\ln(1-\ln(x))+1-\ln(x)}\right)\,\mathrm{d}x\\ &= \int_0^1 \ln(x)\ln(1-\ln(x)) + \ln(x) - \ln^2(x)\,\mathrm{d}x\\ &= \int_{-\infty}^0 e^y y\ln(1-y) + ye^y - y^2e^y\,\mathrm{d}y\\ &= \int_0^\infty -ye^{-y}\ln(1+y) - ye^{-y} - y^2e^{-y}\,\mathrm{d}y\\ &= -4,\end{aligned}$$

the last step obtained through IBP and gamma functions. :)

6. $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\tan(\sin^2(x))}{1-\tan(\sin^2(x))\tan(\cos^2(x))}\,\mathrm{d}x$ and $\displaystyle\int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos(x)+\sin(x)}{\sqrt{2}\sin(x)}\right)+\ln(\tan(x))\,\mathrm{d}x$

Solution. First integral: Apply King's rule to get

$$I := \int_0^{\frac{\pi}{2}} \frac{\tan(\sin^2(x))}{1-\tan(\sin^2(x))\tan(\cos^2(x))}\,\mathrm{d}x = \int_0^{\frac{\pi}{2}} \frac{\tan(\cos^2(x))}{1-\tan(\sin^2(x))\tan(\cos^2(x))}\,\mathrm{d}x.$$

Then we get

$$\begin{aligned} I &= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{\tan(\sin^2(x))+\tan(\cos^2(x))}{1-\tan(\sin^2(x))\tan(\cos^2(x))}\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^{\frac{\pi}{2}} \tan(\sin^2(x)+\cos^2(x))\,\mathrm{d}x \\ &= \frac{\pi}{4}\tan(1). \end{aligned}$$

Second integral: On the first term we recognise

$$I := \int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos(x)+\sin(x)}{\sqrt{2}\sin(x)}\right)\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos\left(\frac{\pi}{4}-x\right)}{\sin(x)}\right)\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos(x)}{\sin\left(\frac{\pi}{4}-x\right)}\right)\,\mathrm{d}x.$$

Then,

$$\begin{aligned} I &= \frac{1}{2}\int_0^{\frac{\pi}{4}} \ln\left(\frac{\cos\left(\frac{\pi}{4}-x\right)}{\sin(x)}\right)+\ln\left(\frac{\cos(x)}{\sin\left(\frac{\pi}{4}-x\right)}\right)\,\mathrm{d}x \\ &= \frac{1}{2}\int_0^{\frac{\pi}{4}} \ln\left(\cot\left(\frac{\pi}{4}-x\right)\right)+\ln(\cot(x))\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{4}} \ln(\cot(x))\,\mathrm{d}x. \end{aligned}$$

Now we see that the original integral evaluates to $\displaystyle\int_0^{\frac{\pi}{4}} \ln(1)\,\mathrm{d}x = 0.$:)

7. $\displaystyle\int_0^1 \frac{1}{(1+x^2)(1+\frac{\pi}{2}+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x$ and $\displaystyle\int_0^1 \frac{1}{(1+x^2)(1+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x$

Solution. In the first integral, we use $\cot^{-1}(x) = \dfrac{\pi}{2} - \tan^{-1}(x)$ repeatedly as

$$\begin{aligned}
&\int_0^1 \frac{1}{(1+x^2)(1+\frac{\pi}{2}+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x \\
&= \int_0^1 \frac{1}{(1+x^2)(1+\tan^{-1}(x))(1+\cot^{-1}(x))}\,\mathrm{d}x \\
&= \frac{1}{2+\frac{\pi}{2}} \int_0^1 \frac{(1+\tan^{-1}(x))+(1+\cot^{-1}(x))}{(1+x^2)(1+\tan^{-1}(x))(1+\cot^{-1}(x))}\,\mathrm{d}x \\
&= \frac{1}{2+\frac{\pi}{2}} \int_0^1 \frac{1}{(1+x^2)(1+\cot^{-1}(x))} + \frac{1}{(1+x^2)(1+\tan^{-1}(x))}\,\mathrm{d}x \\
&= \frac{1}{2+\frac{\pi}{2}} \int_0^1 \frac{1}{(1+x^2)(1+\frac{\pi}{2}-\tan^{-1}(x))} + \frac{1}{(1+x^2)(1+\tan^{-1}(x))}\,\mathrm{d}x \\
&= \frac{1}{2+\frac{\pi}{2}} \int_0^{\frac{\pi}{4}} \frac{1}{1+\frac{\pi}{2}-u} + \frac{1}{1+u}\,\mathrm{d}x \\
&= \frac{1}{2+\frac{\pi}{2}} \left[\ln(1+u) - \ln\left(1+\frac{\pi}{2}-u\right)\right]_0^{\frac{\pi}{4}} \\
&= \frac{1}{2+\frac{\pi}{2}} \ln\left(1+\frac{\pi}{2}\right).
\end{aligned}$$

The second integral is a straight bash. We have

$$\begin{aligned}
&\int_0^1 \frac{1}{(1+x^2)(1+\tan^{-1}(x)\cot^{-1}(x))}\,\mathrm{d}x \\
&= \int_0^1 \frac{1}{(1+x^2)(1+\tan^{-1}(x)(\frac{\pi}{2}-\tan^{-1}(x))}\,\mathrm{d}x \\
&= \int_0^1 \frac{1}{(1+x^2)(1+\frac{\pi}{2}\tan^{-1}(x)-(\tan^{-1}(x))^2)}\,\mathrm{d}x \\
&= \int_0^{\frac{\pi}{4}} \frac{1}{1+\frac{\pi}{2}u-u^2}\,\mathrm{d}u \\
&= \int_0^{\frac{\pi}{4}} \frac{1}{1+\frac{\pi^2}{16}-\left(u-\frac{\pi}{4}\right)^2}\,\mathrm{d}u \\
&= \left[\frac{1}{\sqrt{1+\frac{\pi^2}{16}}} \tanh^{-1}\left(\frac{u-\frac{\pi}{4}}{\sqrt{1+\frac{\pi^2}{16}}}\right)\right]_0^{\frac{\pi}{4}} \\
&= \frac{4}{\sqrt{16+\pi^2}} \tanh^{-1}\left(\frac{\pi}{\sqrt{16+\pi^2}}\right).
\end{aligned}$$

:)

2 Set 1/Segment 2

1. $\displaystyle\int_0^1 \frac{\ln(x^4+1)}{x^3}\,\mathrm{d}x$ and $\displaystyle\int_{-1}^1 \frac{\ln(x^4+1)}{x^3}\,\mathrm{d}x$

   Solution. First integral needs an explicit antiderivative:

   $$\begin{aligned}\int_0^1 \frac{\ln(x^4+1)}{x^3}\,\mathrm{d}x &= \left[-\frac{\ln(x^4+1)}{2x^2}\right]_0^1 + \int_0^1 \frac{2x}{x^4+1}\,\mathrm{d}x \\ &= -\frac{\ln(2)}{2} + [\tan^{-1}(x^2)]_0^1 \\ &= \frac{\pi}{4} - \frac{\ln(2)}{2}.\end{aligned}$$

   Second integral can be worked the same way, or just note the function is odd so the integral is 0. :)

2. $\displaystyle\int \frac{1}{(x+1)\sqrt{x}+2x}\,\mathrm{d}x$ and $\displaystyle\int \frac{1}{(\sqrt{x}+1)x+2\sqrt{x}}\,\mathrm{d}x$

   Solution. For the first integral, let $y=\sqrt{x}$, then

   $$\begin{aligned}\int \frac{1}{(x+1)\sqrt{x}+2x}\,\mathrm{d}x &= \int \frac{2y}{(y^2+1)y+2y^2}\,\mathrm{d}y \\ &= \int \frac{2}{(y+1)^2}\,\mathrm{d}y \\ &= -\frac{2}{y+1}+C \\ &= -\frac{2}{\sqrt{x}+1}+C.\end{aligned}$$

   For the second integral, also let $y=\sqrt{x}$, then

   $$\begin{aligned}\int \frac{1}{(\sqrt{x}+1)x+2\sqrt{x}}\,\mathrm{d}x &= \int \frac{2y}{(y+1)y^2+2y}\,\mathrm{d}y \\ &= \int \frac{2}{y^2+y+2}\,\mathrm{d}y \\ &= \int \frac{2}{\left(y+\frac{1}{2}\right)^2+\frac{7}{4}}\,\mathrm{d}y \\ &= \frac{4}{\sqrt{7}}\tan^{-1}\left(\frac{2y+1}{\sqrt{7}}\right)+C \\ &= \frac{4}{\sqrt{7}}\tan^{-1}\left(\frac{2\sqrt{x}+1}{\sqrt{7}}\right)+C.\end{aligned}$$

   :)

3. $\displaystyle\int \frac{5x^2+3x-2}{x^3+2x^2}\,\mathrm{d}x$ and $\displaystyle\int \frac{(x+1)^2}{(x+2)(x^2+x+1)}\,\mathrm{d}x$

   Solution. These are exercises in partial fractions. The first integral is

   $$\begin{aligned}\int \frac{5x^2+3x-2}{x^3+2x^2}\,\mathrm{d}x &= \int \frac{3}{x+2}+\frac{2}{x}-\frac{1}{x^2}\,\mathrm{d}x \\ &= 3\ln|x+2|+2\ln|x|+\frac{1}{x}+C.\end{aligned}$$

The second integral is

$$\begin{aligned}\int \frac{(x+1)^2}{(x+2)(x^2+x+1)}\,\mathrm{d}x &= \frac{1}{3}\int \frac{2x+1}{x^2+x+1}+\frac{1}{x+2}\,\mathrm{d}x\\ &= \frac{1}{3}(\ln(x^2+x+1)+\ln|x+2|)+C.\end{aligned}$$

Alternatively,

$$\int \frac{(x+1)^2}{(x+2)(x^2+x+1)}\,\mathrm{d}x = \int \frac{(x+1)^2}{1+(x+1)^3}\,\mathrm{d}x = \frac{1}{3}\ln|1+(x+1)^3|+C.$$

:)

4. $\displaystyle\int_0^\infty \frac{1}{(x+\sqrt{x^2+1})(1+x^2)}\,\mathrm{d}x$ and $\displaystyle\int_0^\infty \frac{1}{(x+\sqrt{x^2+1})(1+x)}\,\mathrm{d}x$

Solution. We rationalise the integrals by multiplying numerator and denominator by $\sqrt{x^2+1}-x$:

$$\int_0^\infty \frac{\sqrt{x^2+1}-x}{x^2+1}\,\mathrm{d}x \qquad \text{and} \qquad \int_0^\infty \frac{\sqrt{x^2+1}-x}{x+1}\,\mathrm{d}x.$$

The first integral evaluates as

$$\begin{aligned}\int_0^\infty \frac{\sqrt{x^2+1}-x}{x^2+1}\,\mathrm{d}x &= \int_0^\infty \frac{1}{\sqrt{x^2+1}}-\frac{x}{x^2+1}\,\mathrm{d}x\\ &= \left[\sinh^{-1}(x)-\frac{1}{2}\ln(x^2+1)\right]_0^\infty\\ &= \left[\ln\left(\frac{x+\sqrt{x^2+1}}{\sqrt{x^2+1}}\right)\right]_0^\infty\\ &= \ln(2).\end{aligned}$$

The second integral is much more complicated. We have

$$\int_0^\infty \frac{\sqrt{x^2+1}-x}{x+1}\,\mathrm{d}x = \int_0^\infty \frac{\sqrt{x^2+1}}{x+1}-1+\frac{1}{x+1}\,\mathrm{d}x.$$

Let $x = \tan(u)$, and using Weierstrass substitution after, we have

$$\begin{aligned}
&\int_0^\infty \frac{\sqrt{x^2+1}}{x+1} - 1 + \frac{1}{x+1}\,\mathrm{d}x \\
&= \int_0^{\frac{\pi}{2}} \frac{\sec^3(u)}{\tan(u)+1} - \sec^2(u) + \frac{\sec^2(u)}{\tan(u)+1}\,\mathrm{d}u \\
&= \int_0^1 \frac{\left(\frac{1+t^2}{1-t^2}\right)^3}{\frac{2t}{1-t^2}+1}\cdot\frac{2}{1+t^2} - \left(\frac{1+t^2}{1-t^2}\right)^2\cdot\frac{2}{1+t^2} + \frac{\left(\frac{1+t^2}{1-t^2}\right)^2}{\frac{2t}{1-t^2}+1}\cdot\frac{2}{1+t^2}\,\mathrm{d}t \\
&= \int_0^1 \frac{2(1+t^2)^2}{(1-t^2)^2(2t+1-t^2)} - \frac{2(1+t^2)}{(1-t^2)^2} + \frac{2(1+t^2)}{(1-t^2)(2t+1-t^2)}\,\mathrm{d}t \\
&= \int_0^1 \frac{2(1+t^2)^2 - 2(1+t^2)(2t+1-t^2) + 2(1-t^4)}{(1-t^2)^2(2t+1-t^2)}\,\mathrm{d}t \\
&= 2\int_0^1 \frac{(1+t^2)(1-t)^2}{(1-t^2)^2(2t+1-t^2)}\,\mathrm{d}t \\
&= 2\int_0^1 \frac{(1+t^2)}{(1+t)^2(2t+1-t^2)}\,\mathrm{d}t \\
&= \text{partial fractions galore} \\
&= 2\left[\frac{1}{1+t} - \ln|1+t| + \frac{\sqrt{2}+1}{2}\ln|\sqrt{2}-1+t| - \frac{\sqrt{2}-1}{2}\ln|\sqrt{2}+1-t|\right]_0^1 \\
&= -1 - \ln(2) - (\sqrt{2}+1)\ln(\sqrt{2}-1) + (\sqrt{2}-1)\ln(\sqrt{2}+1).
\end{aligned}$$

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5. $\displaystyle\int_0^1 \ln\left(1+\sqrt{1+\left(\frac{x^4-1}{2x^2}\right)^2}\right)\,\mathrm{d}x$ and $\displaystyle\int_0^\infty \exp\left(-1-\sqrt{1+\left(\frac{x^4-1}{2x^2}\right)^2}\right)\,\mathrm{d}x$

Solution. We simplify the funky thing first:

$$\begin{aligned}
1+\sqrt{1+\left(\frac{x^4-1}{2x^2}\right)^2} &= 1+\sqrt{\left(\frac{x^4+1}{2x^2}\right)^2} \\
&= 1+\frac{x^4+1}{2x^2} \\
&= \frac{1}{2}\left(\frac{x^2+1}{x}\right)^2 \\
&= \frac{1}{2}\left(x+\frac{1}{x}\right)^2.
\end{aligned}$$

Thus we are looking at the integrals

$$-\ln(2) + 2\int_0^1 \ln\left(x+\frac{1}{x}\right)\,\mathrm{d}x \qquad \text{and} \qquad \int_0^\infty e^{-\frac{1}{2}\left(x+\frac{1}{x}\right)^2}\,\mathrm{d}x.$$

The logarithmic integral is evaluated by parts as

$$\begin{aligned}\int_0^1 \ln\left(x+\frac{1}{x}\right)\,\mathrm{d}x &= \left[x\ln\left(x+\frac{1}{x}\right)\right]_0^1 - \int_0^1 \frac{x^2-1}{x^2+1}\,\mathrm{d}x\\ &= \left[x\ln\left(x+\frac{1}{x}\right) - x + 2\tan^{-1}(x)\right]_0^1\\ &= \ln(2) - 1 + \frac{\pi}{2}.\end{aligned}$$

Thus the answer is $\ln(2)+\pi-2$.

The exponential integral is integrated first by noting that

$$\int_0^\infty e^{-\frac{1}{2}\left(x+\frac{1}{x}\right)^2}\,\mathrm{d}x = \int_{-\infty}^0 e^{-\frac{1}{2}\left(x+\frac{1}{x}\right)^2}\,\mathrm{d}x,$$

then by Glassers master theorem we have

$$\begin{aligned}\int_0^\infty e^{-\frac{1}{2}\left(x+\frac{1}{x}\right)^2}\,\mathrm{d}x &= \frac{1}{2e^2}\int_{-\infty}^\infty e^{-\frac{1}{2}\left(x-\frac{1}{x}\right)^2}\,\mathrm{d}x\\ &= \frac{1}{2e^2}\int_{-\infty}^\infty e^{-\frac{1}{2}x^2}\,\mathrm{d}x\\ &= \frac{\sqrt{2\pi}}{2e^2}.\end{aligned}$$

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6. $\displaystyle\int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\ln(x)\,\mathrm{d}x$ and $\displaystyle\int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\,\mathrm{d}x$

Solution. The first integral evaluates as

$$\begin{aligned}I &= \int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\ln(x)\,\mathrm{d}x\\ &= \int_0^\infty \frac{1-\cos^2(x)}{x^2}\ln(x)\,\mathrm{d}x - \int_0^\infty \frac{1-\cos(x)}{x^2}\ln(x)\,\mathrm{d}x\\ &= \int_0^\infty \frac{\sin^2(x)}{x^2}\ln(x)\,\mathrm{d}x - \int_0^\infty \frac{2\sin^2\left(\frac{x}{2}\right)}{x^2}\ln(x)\,\mathrm{d}x\\ &= \int_0^\infty \frac{\sin^2(x)}{x^2}\ln(x)\,\mathrm{d}x - \int_0^\infty \frac{\sin^2(x)}{x^2}\ln(2x)\,\mathrm{d}x\\ &= -\ln(2)\int_0^\infty \frac{\sin^2(x)}{x^2}\,\mathrm{d}x\\ &= -\frac{\pi\ln(2)}{2}.\end{aligned}$$

The second integral evaluates as

$$\begin{aligned}I &= \int_0^\infty \frac{\cos(x)-\cos^2(x)}{x^2}\,\mathrm{d}x\\ &= \int_0^\infty \frac{1-\cos^2(x)}{x^2}\,\mathrm{d}x - \int_0^\infty \frac{1-\cos(x)}{x^2}\,\mathrm{d}x\\ &= \int_0^\infty \frac{\sin^2(x)}{x^2}\,\mathrm{d}x - \int_0^\infty \frac{2\sin^2\left(\frac{x}{2}\right)}{x^2}\,\mathrm{d}x\\ &= \int_0^\infty \frac{\sin^2(x)}{x^2}\,\mathrm{d}x - \int_0^\infty \frac{\sin^2(x)}{x^2}\,\mathrm{d}x\\ &= 0.\end{aligned}$$

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7. $\lim_{n\to\infty}\int_1^n \ln\left(1+\frac{e}{n}+\frac{1}{\sqrt[e]{nx}}\right)\,\mathrm{d}x$ and $\lim_{n\to\infty}\int_1^n \ln\left(1+\frac{2}{n}+\frac{1}{\sqrt{nx}}\right)\,\mathrm{d}x$

Solution. We consider the general form

$$\int_1^n \ln\left(a_{n,p}+\frac{1}{\sqrt[p]{nx}}\right)\,\mathrm{d}x$$

and integrate by putting $x=\dfrac{1}{ny^p}$:

$$\begin{aligned}
&\frac{1}{n}\int_{\frac{1}{\sqrt[p]{n^2}}}^{\frac{1}{\sqrt[p]{n}}} \frac{p}{y^{p+1}}\ln\left(a_{n,p}+y\right)\,\mathrm{d}y\\
&\quad=\left[-\frac{\ln\left(a_{n,p}+y\right)}{ny^p}\right]_{\frac{1}{\sqrt[p]{n^2}}}^{\frac{1}{\sqrt[p]{n}}}+\frac{1}{n}\int_{\frac{1}{\sqrt[p]{n^2}}}^{\frac{1}{\sqrt[p]{n}}}\frac{1}{y^p\left(a_{n,p}+y\right)}\,\mathrm{d}y\\
&\quad=n\ln\left(a_{n,p}+\frac{1}{\sqrt[p]{n^2}}\right)-\ln\left(a_{n,p}+\frac{1}{\sqrt[p]{n}}\right)+\frac{1}{n}\int_{\frac{1}{\sqrt[p]{n^2}}}^{\frac{1}{\sqrt[p]{n}}}\frac{1}{y^p\left(a_{n,p}+y\right)}\,\mathrm{d}y.
\end{aligned}$$

In the case $a_{n,p}=1+\dfrac{e}{n}$,

$$\underbrace{n\ln\left(1+\frac{e}{n}+\frac{1}{\sqrt[e]{n^2}}\right)}_{\sim n^{1-\frac{2}{e}}\to+\infty}-\underbrace{\ln\left(1+\frac{e}{n}+\frac{1}{\sqrt[e]{n}}\right)}_{\to 0}+\underbrace{\frac{1}{n}\int_{\frac{1}{\sqrt[e]{n^2}}}^{\frac{1}{\sqrt[e]{n}}}\frac{1}{y^e\left(1+\frac{e}{n}+y\right)}\,\mathrm{d}y}_{>0}$$

so the limit is $+\infty$.

In the case $a_{n,p}=1+\dfrac{2}{n}$,

$$\underbrace{n\ln\left(1+\frac{2}{n}+\frac{1}{n}\right)}_{\to 3}-\underbrace{\ln\left(1+\frac{2}{n}+\frac{1}{\sqrt{n}}\right)}_{\to 0}+\underbrace{\frac{1}{n}\int_{\frac{1}{n}}^{\frac{1}{\sqrt{n}}}\frac{1}{y^2\left(1+\frac{2}{n}+y\right)}\,\mathrm{d}y}_{>0}$$

so we need to do some extra work. The integral

$$\begin{aligned}
&\frac{1}{n}\int_{\frac{1}{n}}^{\frac{1}{\sqrt{n}}}\frac{1}{y^2\left(a_{n,p}+y\right)}\,\mathrm{d}y\\
&\quad=\frac{1}{n}\int_{\frac{1}{n}}^{\frac{1}{\sqrt{n}}}\frac{1}{a_{n,p}y^2}-\frac{1}{a_{n,p}^2y}+\frac{1}{a_{n,p}^2(y+a_{n,p})}\,\mathrm{d}y\\
&\quad=\frac{1}{n}\left[\frac{\ln(y+a_{n,p})-\ln(y)}{a_{n,p}^2}-\frac{1}{a_{n,p}y}\right]_{\frac{1}{n}}^{\frac{1}{\sqrt{n}}}\\
&\quad=\underbrace{\frac{\ln\left(\frac{1}{\sqrt{n}}+1+\frac{2}{n}\right)-\ln\left(\frac{1}{\sqrt{n}}\right)}{n\left(1+\frac{2}{n}\right)^2}}_{\to 0}-\underbrace{\frac{1}{\left(1+\frac{2}{n}\right)\sqrt{n}}}_{\to 0}-\underbrace{\frac{\ln\left(\frac{1}{n}+1+\frac{2}{n}\right)-\ln\left(\frac{1}{n}\right)}{n\left(1+\frac{2}{n}\right)^2}}_{\to 0}+\underbrace{\frac{1}{\left(1+\frac{2}{n}\right)}}_{\to 1}\\
&\quad\to 1.
\end{aligned}$$

Hence the limit is $3+0+1=4$. :)

3 Set 3/Segment 3

1. $\displaystyle\int_0^1 \frac{x^4+1}{x^6+1}\,\mathrm{d}x$ and $\displaystyle\int_0^1 \frac{x^2+1}{x^3+1}\,\mathrm{d}x$

   Solution. The challenge here is simple: figure out which simplifies nicer. In the first integral, observe that $x^6+1=(x^2+1)(x^4-x^2+1)$, so that

   $$\int_0^1 \frac{x^4+1}{x^6+1}\,\mathrm{d}x=\int_0^1 \frac{1}{x^2+1}+\frac{x^2}{x^6+1}\,\mathrm{d}x=\left[\tan^{-1}(x)+\frac{1}{3}\tan^{-1}(x^3)\right]_0^1=\frac{\pi}{3}.$$

   The second integral doesn't work the same way, since the denominator is not of even power. Expanding into partial fractions yields

   $$\begin{aligned}\int_0^1 \frac{x^2+1}{x^3+1}\,\mathrm{d}x&=\frac{1}{3}\int_0^1 \frac{2}{x+1}+\frac{x+1}{x^2-x+1}\,\mathrm{d}x\\&=\frac{1}{3}\int_0^1 \frac{2}{x+1}+\frac{x-\frac{1}{2}}{x^2-x+1}+\frac{\frac{3}{2}}{\left(x-\frac{1}{2}\right)^2+\frac{3}{4}}\,\mathrm{d}x\\&=\frac{1}{3}\left[2\ln|x+1|+\frac{1}{2}\ln|x^2-x+1|+\sqrt{3}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)\right]_0^1\\&=\frac{1}{3}\left(2\ln(2)+\frac{\sqrt{3}\pi}{3}\right).\end{aligned}$$

   :)

2. $\displaystyle\int_0^\infty \frac{1}{\sqrt{1+\sqrt{e^x}}}\,\mathrm{d}x$ and $\displaystyle\int_1^\infty e^{-\sqrt{\sqrt{x}-1}}\,\mathrm{d}x$

   Solution. Both of these are forced substitutions. In the first integral let $u=\sqrt{1+\sqrt{e^x}}$. Then $u'=\dfrac{1}{2u}\cdot\dfrac{1}{2\sqrt{e^x}}\cdot e^x=\dfrac{u^2-1}{4u}$. So we have

   $$\begin{aligned}\int_0^\infty \frac{1}{\sqrt{1+\sqrt{e^x}}}\,\mathrm{d}x&=\int_{\sqrt{2}}^\infty \frac{4}{u^2-1}\,\mathrm{d}u\\&=2[\ln|u-1|-\ln|u+1|]_{\sqrt{2}}^\infty\\&=2(\ln(\sqrt{2}+1)-\ln(\sqrt{2}-1)).\end{aligned}$$

   In the second integral let $u=\sqrt{\sqrt{x}-1}$. Then $u'=\dfrac{1}{2u}\cdot\dfrac{1}{2\sqrt{x}}=\dfrac{1}{4u(1+u^2)}$. It follows that

   $$\begin{aligned}\int_1^\infty e^{-\sqrt{\sqrt{x}-1}}\,\mathrm{d}x&=\int_0^\infty 4u(1+u^2)e^{-u}\,\mathrm{d}u\\&=4\left(\int_0^\infty ue^{-u}\,\mathrm{d}u+\int_0^\infty u^3e^{-u}\,\mathrm{d}u\right)\\&=28.\end{aligned}$$

   :)

3. $\displaystyle\int_0^\infty \frac{\sqrt[3]{x}}{1+x^3}\,\mathrm{d}x$ vs $\displaystyle\int_0^\infty \frac{\sqrt[3]{x}}{1+x^4}\,\mathrm{d}x$

Solution. Both problems can be solved using power substitution into beta function, with

$$\begin{aligned}\int_0^\infty \frac{x^a}{1+x^b}\,\mathrm{d}x &= \frac{1}{b}\int_0^\infty \frac{y^{\frac{a+1}{b}-1}}{1+y}\,\mathrm{d}y \\ &= \frac{1}{b}\,\mathrm{B}\left(\frac{a+1}{b}, 1-\frac{a+1}{b}\right) \\ &= \frac{1}{b}\Gamma\left(\frac{a+1}{b}\right)\Gamma\left(1-\frac{a+1}{b}\right) \\ &= \frac{\pi}{b}\csc\left(\frac{a+1}{b}\pi\right).\end{aligned}$$

The first integral gives

$$\frac{\pi}{3}\csc\left(\frac{\frac{1}{3}+1}{3}\pi\right) = \frac{\pi}{3}\csc\left(\frac{4\pi}{9}\right).$$

The second integral gives

$$\frac{\pi}{4}\csc\left(\frac{\frac{1}{3}+1}{4}\pi\right) = \frac{\pi}{4}\csc\left(\frac{\pi}{3}\right) = \frac{\pi}{2\sqrt{3}}.$$

:)

4. $\displaystyle\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x^2}\right)}{x}\,\mathrm{d}x$ and $\displaystyle\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x}\right)}{x^2}\,\mathrm{d}x$

Solution. The first integral is just Taylor expansion:

$$\begin{aligned}\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x^2}\right)}{x}\,\mathrm{d}x &= \sum_{n=0}^\infty \int_1^\infty \frac{(-1)^n}{(2n+1)x^{2(2n+1)}}\cdot\frac{1}{x}\,\mathrm{d}x \\ &= \sum_{n=0}^\infty \int_1^\infty \frac{(-1)^n}{(2n+1)x^{4n+3}}\,\mathrm{d}x \\ &= \sum_{n=0}^\infty \left[\frac{(-1)^{n+1}}{(4n+2)(2n+1)x^{4n+2}}\right]_1^\infty \\ &= \frac{1}{2}\sum_{n=0}^\infty \frac{(-1)^n}{(2n+1)^2} \\ &= \frac{G}{2}.\end{aligned}$$

The second integral uses the substitution $y = \dfrac{1}{x}$:

$$\begin{aligned}\int_1^\infty \frac{\tan^{-1}\left(\frac{1}{x}\right)}{x^2}\,\mathrm{d}x &= \int_0^1 \tan^{-1}(y)\;\mathrm{d}y \\ &= \left[y\tan^{-1}(y)\right]_0^1 - \int_0^1 \frac{y}{y^2+1}\,\mathrm{d}y \\ &= \frac{\pi}{4} - \left[\frac{1}{2}\ln(y^2+1)\right]_0^1 \\ &= \frac{\pi}{4} - \frac{\ln(2)}{2}.\end{aligned}$$

:)

5. $\displaystyle\int_0^\infty \frac{\sqrt[3]{(1+\frac{3}{x})^x} - \sqrt{(1+\frac{2}{x})^x}}{x}\,\mathrm{d}x$ and $\displaystyle\int_{-\infty}^\infty \sin(x^2 + x\sqrt{3\pi} + \pi)\,\mathrm{d}x$

Solution. The first integral is a straight Frullani integral as

$$\int_0^\infty \frac{\sqrt[3]{(1+\frac{3}{x})^x} - \sqrt{(1+\frac{2}{x})^x}}{x}\,\mathrm{d}x = \int_0^\infty \frac{f(\frac{1}{3}x) - f(\frac{1}{2}x)}{x}\,\mathrm{d}x$$

with $f(x) = \left(1 + \dfrac{1}{x}\right)^x$. The value is $(e-1)\ln\left(\dfrac{2}{3}\right)$.

The second integral, unfortunately, has no such nice solution. Completing the square we have

$$\begin{aligned}
\int_{-\infty}^\infty \sin(x^2 + x\sqrt{3\pi} + \pi)\,\mathrm{d}x &= \int_{-\infty}^\infty \sin\left(\left(x + \frac{\sqrt{3\pi}}{2}\right)^2 + \frac{\pi}{4}\right)\,\mathrm{d}x \\
&= \int_{-\infty}^\infty \sin\left(x^2 + \frac{\pi}{4}\right)\,\mathrm{d}x \\
&= 2\int_0^\infty \sin\left(x^2 + \frac{\pi}{4}\right)\,\mathrm{d}x \\
&= \sqrt{2}\int_0^\infty \sin(x^2) + \cos(x^2)\,\mathrm{d}x \\
&= 2\sqrt{2}\int_0^\infty \sin(x^2)\,\mathrm{d}x \\
&= \sqrt{\pi}.
\end{aligned}$$

:)

6. $\displaystyle\int_0^{\frac{\pi}{4}} \cos^8(x) - \sin^8(x)\,\mathrm{d}x$ and $\displaystyle\int_0^{\frac{\pi}{4}} \cos^6(x) - \sin^6(x)\,\mathrm{d}x$

Solution. Both integrals are an exercise in factorisation. For the first integral, we have

$$\begin{aligned}
\int_0^{\frac{\pi}{4}} \cos^8(x) - \sin^8(x)\,\mathrm{d}x &= \int_0^{\frac{\pi}{4}} (\cos^4(x) + \sin^4(x))(\cos^2(x) - \sin^2(x))\,\mathrm{d}x \\
&= \int_0^{\frac{\pi}{4}} \cos(2x)(1 - 2\sin^2(x)\cos^2(x))\,\mathrm{d}x \\
&= \int_0^{\frac{\pi}{4}} \cos(2x)\left(1 - \frac{\sin^2(2x)}{2}\right)\,\mathrm{d}x \\
&= \left[\frac{\sin(2x)}{2} - \frac{\sin^3(2x)}{12}\right]_0^{\frac{\pi}{4}} \\
&= \frac{5}{12}.
\end{aligned}$$

For the second integral,

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \cos^6(x) - \sin^6(x)\,\mathrm{d}x &= \int_0^{\frac{\pi}{4}} (\cos^2(x) - \sin^2(x))(\cos^4(x) + \sin^4(x) + \sin^2(x)\cos^2(x))\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{4}} \cos(2x)(1 - \sin^2(x)\cos^2(x))\,\mathrm{d}x \\ &= \int_0^{\frac{\pi}{4}} \cos(2x)\left(1 - \frac{\sin^2(2x)}{4}\right)\,\mathrm{d}x \\ &= \left[\frac{\sin(2x)}{2} - \frac{\sin^3(2x)}{24}\right]_0^{\frac{\pi}{4}} \\ &= \frac{11}{24}.\end{aligned}$$

:)

7. $\displaystyle\int_0^1 \sqrt{\frac{x}{1-x}}\ln\left(\frac{x}{1-x}\right)\,\mathrm{d}x$ and $\displaystyle\int_0^1 \frac{\ln(\sqrt{x}+\sqrt{1-x})}{\sqrt{x}}\,\mathrm{d}x$

Solution. Let $x = \sin^2(u)$. Then we have

$$\begin{aligned}\int_0^1 \sqrt{\frac{x}{1-x}}\ln\left(\frac{x}{1-x}\right)\,\mathrm{d}x &= \int_0^{\frac{\pi}{2}} \tan(u)\ln(\tan^2(u))\cdot 2\sin(u)\cos(u)\,\mathrm{d}u \\ &= 4\int_0^{\frac{\pi}{2}} \sin^2(u)\ln(\tan(u))\,\mathrm{d}u \\ &= -4\int_0^{\frac{\pi}{2}} \cos^2(u)\ln(\tan(u))\,\mathrm{d}u,\end{aligned}$$

the last step by King's rule. Letting I be the original integral, it follows

$$\begin{aligned}I &= 2\int_0^{\frac{\pi}{2}} (\sin^2(u) - \cos^2(u))\ln(\tan(u))\,\mathrm{d}u \\ &= -2\int_0^{\frac{\pi}{2}} \cos(2u)\ln(\tan(u))\,\mathrm{d}u \\ &= -2\left(\left[\frac{\sin(2u)\ln(\tan(u))}{2}\right]_0^{\frac{\pi}{2}} - \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin(2u)\frac{\sec^2(u)}{\tan(u)}\,\mathrm{d}u\right) \\ &= 2\int_0^{\frac{\pi}{2}} 1\,\mathrm{d}u \\ &= \pi.\end{aligned}$$

Second integral: Again let $x = \sin^2(u)$. Then we have

$$\begin{aligned}\int_0^1 \frac{\ln(\sqrt{x}+\sqrt{1-x})}{\sqrt{x}}\,\mathrm{d}x &= 2\int_0^{\frac{\pi}{2}} \cos(u)\ln(\sin(u)+\cos(u))\,\mathrm{d}u \\ &= 2\int_0^{\frac{\pi}{2}} \sin(u)\ln(\sin(u)+\cos(u))\,\mathrm{d}u,\end{aligned}$$

the last line following from King's rule. Letting the original integral be I, we get

$$\begin{aligned}
I &= \int_0^{\frac{\pi}{2}} (\sin(u) + \cos(u)) \ln(\sin(u) + \cos(u)) \, \mathrm{d}u \\
&= \sqrt{2} \int_0^{\frac{\pi}{2}} \cos\left(u - \frac{\pi}{4}\right) \ln\left(\sqrt{2}\cos\left(u - \frac{\pi}{4}\right)\right) \, \mathrm{d}u \\
&= \sqrt{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos(u) \ln(\sqrt{2}\cos(u)) \, \mathrm{d}u \\
&= \frac{\sqrt{2}\ln(2)}{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos(u) \, \mathrm{d}u + \sqrt{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos(u) \ln(\cos(u)) \, \mathrm{d}u \\
&= \ln(2) + \sqrt{2} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos(u) \ln(\cos(u)) \, \mathrm{d}u.
\end{aligned}$$

This last integral is evaluated by parts, as

$$\begin{aligned}
\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos(u) \ln(\cos(u)) \, \mathrm{d}u &= \left[\sin(u) \ln(\cos(u))\right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}} - \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\sin^2(u)}{\cos(u)} \, \mathrm{d}u \\
&= -\frac{\sqrt{2}\ln(2)}{2} + 2\int_0^{\frac{1}{\sqrt{2}}} \frac{y^2}{1-y^2} \, \mathrm{d}y \\
&= \text{Partial fractions} \\
&= -\frac{\sqrt{2}\ln(2)}{2} - \sqrt{2} + 2\ln(\sqrt{2}+1).
\end{aligned}$$

Therefore we get $I = 2\sqrt{2}\ln(\sqrt{2}+1) - 2$. :)